

Name	Subject Class	Class	Candidate Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2016

BIOLOGY

HIGHER 2

9648/02
22 AUGUST 2016
2 hours

Paper 2 Core Paper

Additional Material: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class on this answer booklet.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, circle the number of the Section B question you have answered in the grid opposite.
Fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
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8	
Section B	
9 or 10	
Total	100

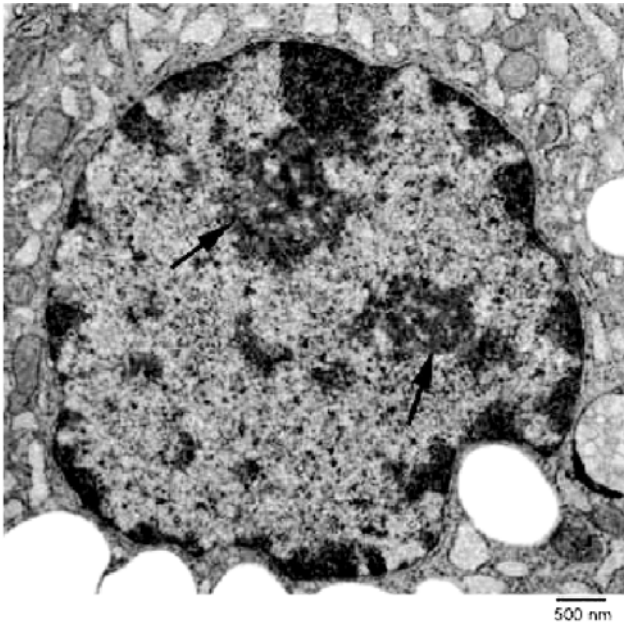
This question paper consists of **21** printed pages.

[Turn over

- 1 The appearance of cancer cells has been known to be different from normal cells. These differences have been used as a method of diagnosis by doctors. Prostate cells taken from wild-type mice and mice with prostate cancer were analysed.

Fig. 1.1 compares the same organelle found in these cells viewed under the electron microscope, while Fig. 1.2 shows the levels of ribosomal RNA (rRNA) measured in these two types of cells.

Part of a prostate cell from wild-type mice



Part of a prostate cell from mice with prostate cancer

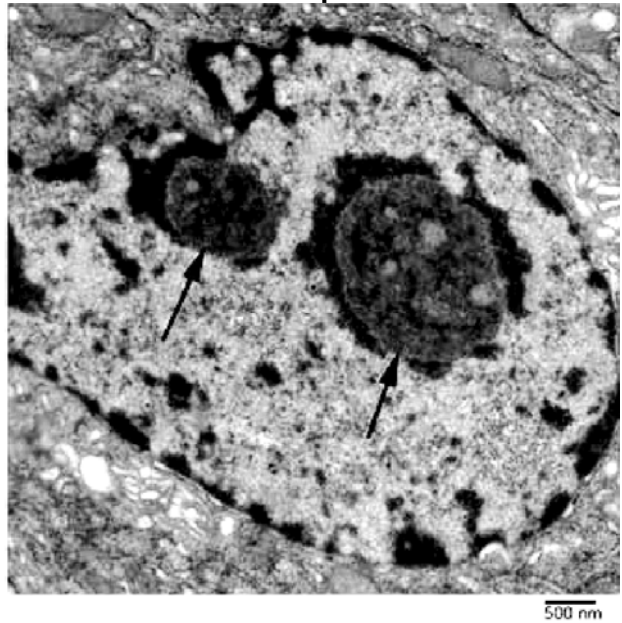


Fig. 1.1

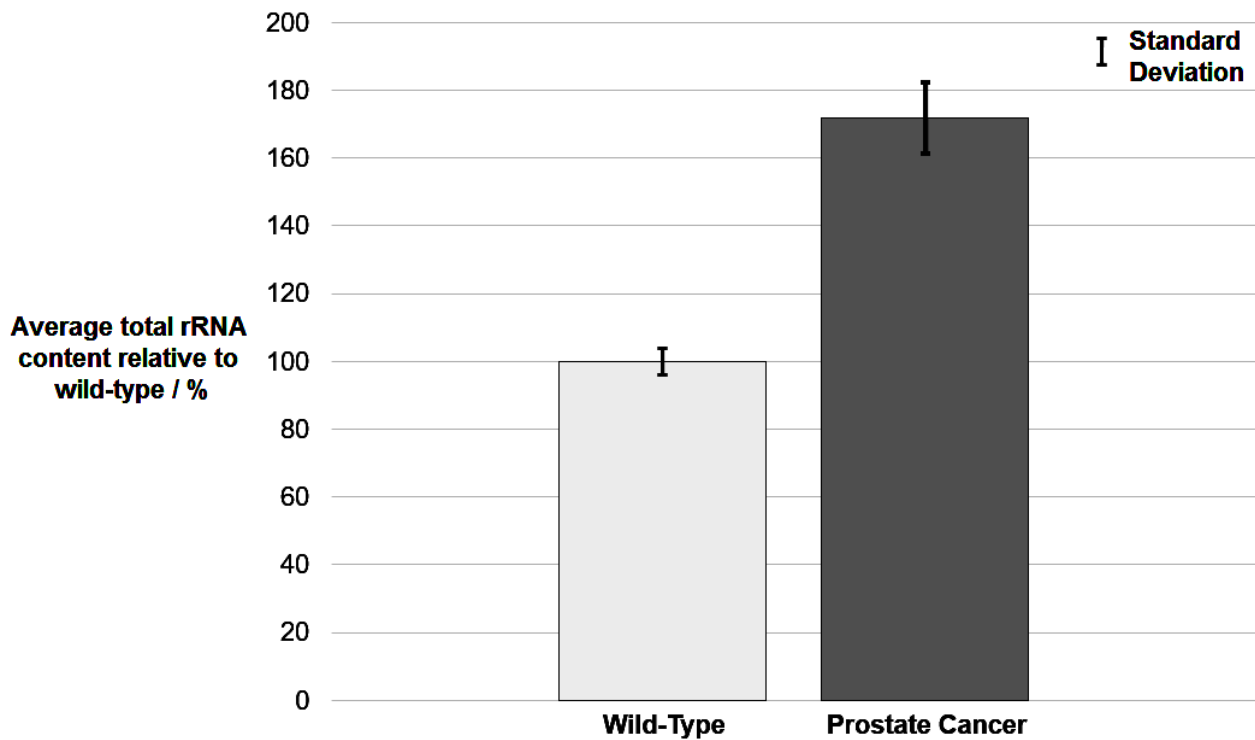


Fig. 1.2

(a) (i) Identify the regions indicated by the arrows in Fig. 1.1.

Nucleoli;

R! Nucleolus [1]

(ii) With reference to the regions identified in Fig. 1.1 and information given in Fig. 1.2, describe the differences between wild-type prostate cells and prostate cancer cells.

- 1. Prostate cancer cells have denser / larger nucleoli;**
- 2. Prostate cancer cells contain more rRNA / show greater variation in rRNA content compared to wild-type cells + Ref. to data from Fig. 1.2 (72% higher in average total rRNA content / higher standard deviation of 20% vs 8%);**

@ 1m

[2]

(iii) Explain the differences described in (a)(ii).

- 1. Nucleoli are the sites of rRNA synthesis through transcription / sites of assembly of ribosomal subunits, hence appear denser in cancer cells;**
- 2. Cancer cells undergo higher rates of protein synthesis in preparation for faster rate of cell division;**
- 3. Higher rates of translation is facilitated by a greater number of ribosomes, hence the higher rRNA content in cancer cells;**
- 4. (Different cancer cells have different mutations resulting in) dysregulation of gene expression in cancer cells lead to different cells having different rates of protein synthesis and greater variation in rRNA content;**

@ 1m, max 3

[3]

Studies on ribosomal proteins have shown that these proteins undergo a high degree of post-translational modifications, including methylation, acetylation and phosphorylation.

(b) Suggest reasons why post-translational modification of ribosomal proteins is needed.

- 1. Plays a role in the regulation of rate of translation / rate of protein synthesis / translation accuracy;**
- 2. Affects the folding / 3D configuration of ribosomal proteins, which affects their binding with rRNA / assembly of ribosomal subunits;**
- 3. Contributes to the stability of ribosomal proteins, making them more resistant to degradation by proteases;**
- 4. Provides for diversity in protein structure beyond that allowed by the 20 encoded amino acids;**
- 5. Ref. to inactive proteins becoming active, or making proteins functional;**

@ 1m, max 2

[2]

[Total: 8m]

- 2 The meiotic cell cycle in the diploid germ cells of an organism can be followed by measuring the number of chromosomes as well as the amount of DNA material per cell over a period of time. Fig. 2.1 shows the results of the analyses, beginning with the start of prophase.

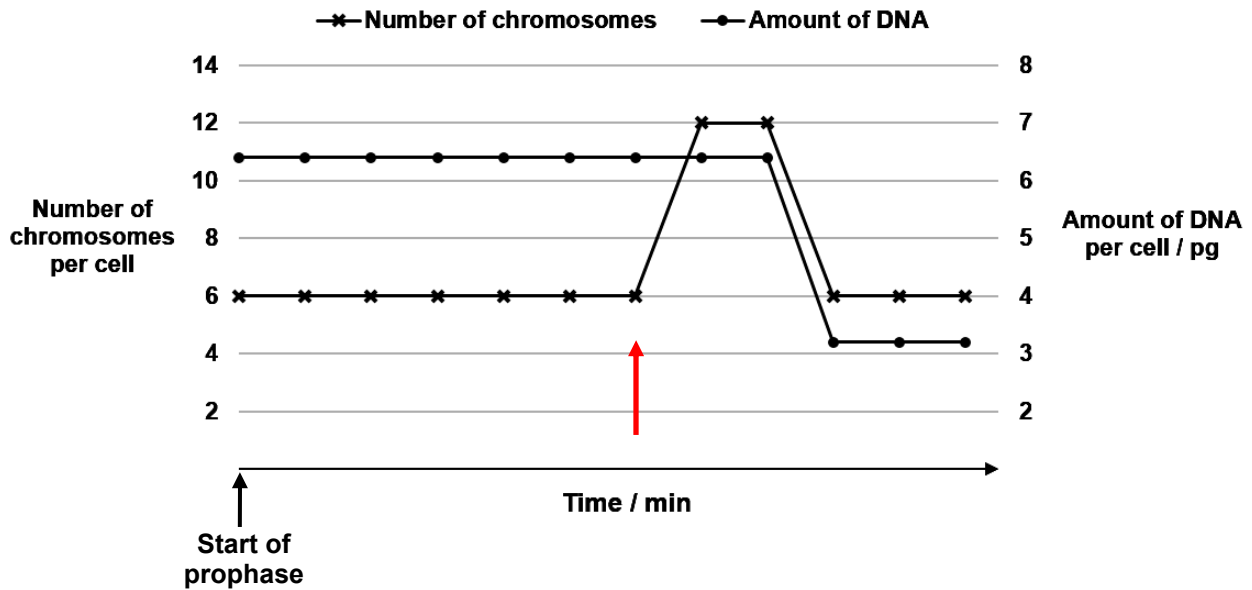


Fig. 2.1

- (a) (i) On Fig. 2.1, indicate with an arrow where anaphase begins.
Indicated as red arrow above; [1]

- (ii) Explain whether Fig. 2.1 shows the first or the second meiotic division.

1. Second meiotic division / Meiosis II;
2. The number of chromosomes at the end of the division remained the same as that during prophase, of 6 chromosomes per cell / Cells were haploid at the start of the division and remained haploid;
3. During anaphase II, chromatids were separated to opposite poles and each chromatid is considered as an individual chromosome (hence no. of chromosomes / ploidy level remain unchanged); OR
The number of chromosomes at the end of meiosis I would have been half that of the parent cell / cells would be diploid at the start of the division but daughter cells would be haploid;

@ 1m

[3]

- (iii) The centromere of a chromosome comprises non-coding tandem repeats. Suggest how the structure of the centromere allows it to carry out one of its functions in cell division.

1. Nucleotide sequence of the centromere results in a specific 3D conformation which is complementary to (DNA-)binding site of proteins;
2. which facilitate adhesion of sister chromatids / which make up the kinetochore complex to allow attachment of spindle fibres;

@ 1m

[2]

In normal cells, the ends of chromosomes do not normally fuse with each other. However, in senescent cells where the ends of chromosomes are eroded to a critical length due to the end-replication problem, chromosomes may undergo end-to-end fusions. Fig. 2.2 shows how the ends of two sister chromatids may fuse and subsequently break during cell division at a location other than the point of fusion.

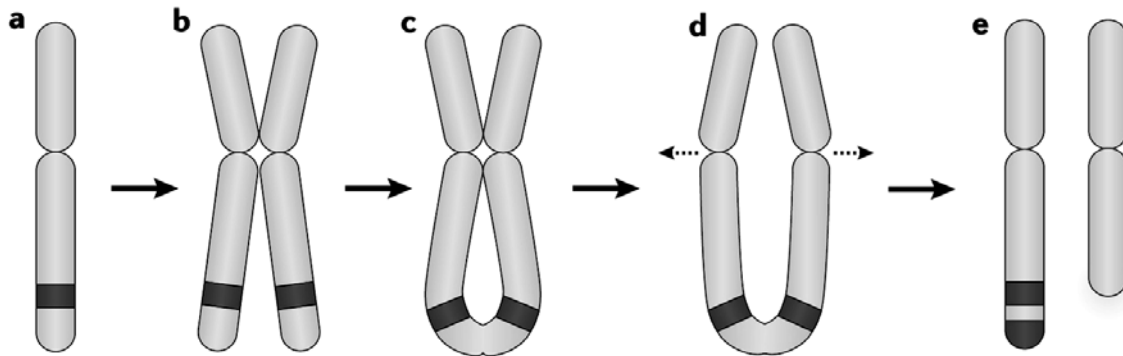


Fig. 2.2

- (6) (i) Describe the result of the chromosomal mutation on the chromosomes shown in Fig. 2.2i.

**A deletion on one chromosome and a duplication on the other chromosome;
R! translocation mutation**

@ 1m

[1]

- (ii) Suggest how the ends of chromosomes are normally protected from end-to-end fusions.

1. Telomeres form loops at the ends of chromosomes, hence there are no free ends for fusion to occur;

2. Proteins bind to telomeres at the ends of chromosomes, preventing the fusion of these ends;

@ 1m, max 1

[1]

- (iii) Explain how the end-replication problem will lead to the erosion of the ends of chromosomes.

1. During the replication of DNA, the RNA primer at the 5' end of the lagging strand is removed;

2. but the gap cannot be filled with complementary deoxyribonucleotides;

3. Due to the absence of an existing 3'-OH group for DNA polymerase I to add nucleotides to / DNA polymerase I can only elongate a strand in a 5' to 3' direction;

4. Hence, over repeated cycles of DNA replication, there will be a gradual shortening of the ends of chromosomes;

@ 1m

[3]

[Total: 11m]

- 3 (a) A new type of influenza drug has been shown to be effective against drug-resistant strains of the flu virus, according to a study led by University of British Columbia researchers in 2013. This drug works by inhibiting newly-formed influenza viruses from leaving the cell surface membrane of host cells, thus preventing the infection of other host cells.

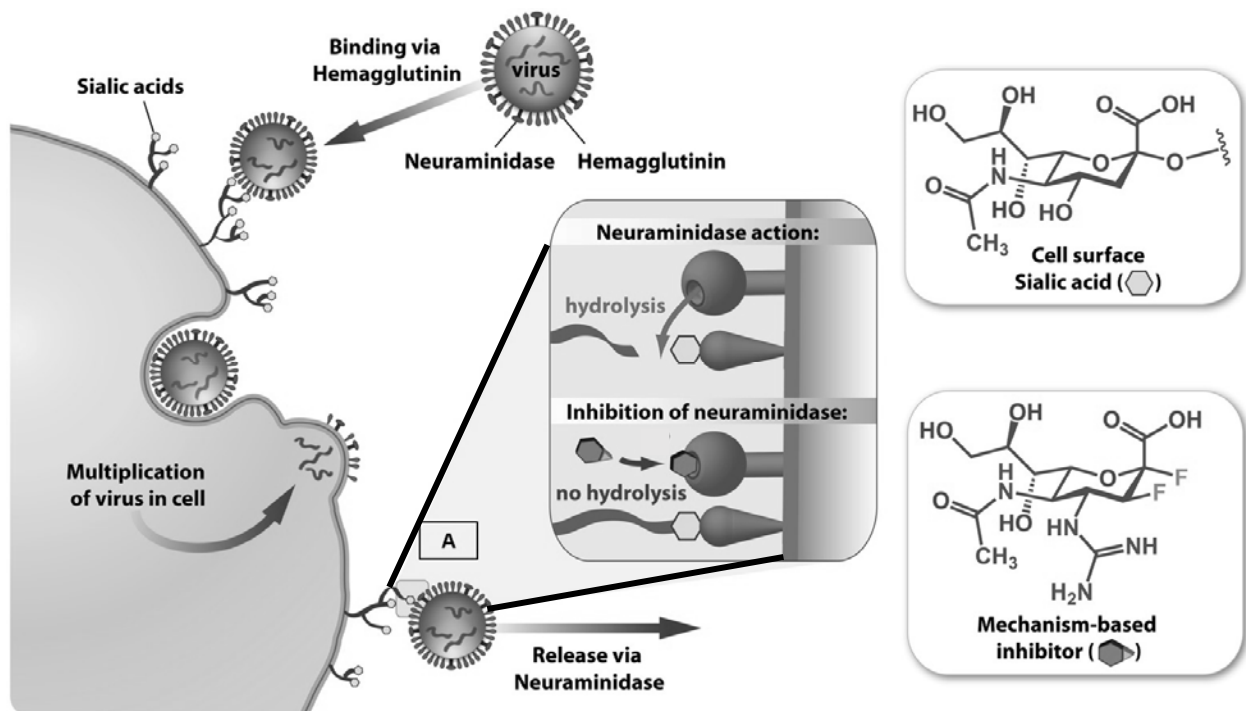


Fig. 3.1

- (i) Describe the process by which influenza viruses replicate in host cells.

1. Viral RNA genome is transcribed to mRNA by viral RNA-dependent RNA polymerase inside nucleus;
2. The mRNA is transported to the cytoplasm and translated to viral structural proteins and viral enzymes by host cell ribosomes;
3. The mRNA in the nucleus is also used as a template to replicate new RNA genomes for new influenza viruses;
4. Capsid proteins are transported back to the nucleus to bind with viral RNA to form ribonucleoproteins (RNPs);
5. Neuraminidase and haemagglutinin are transported through the Golgi apparatus and incorporated onto the cell surface;

@ 1m, max 4 [4]

(ii) With reference to Fig. 3.1, describe how the drug prevents the newly formed influenza viruses from infecting other host cells in the process labelled **A**.

1. The drug is a competitive inhibitor of neuraminidase as it is structurally similar to the cell surface sialic acid;;
2. Hence the drug binds to the complementary active site on neuraminidase instead of sialic acid;
3. This prevents neuraminidase from hydrolysing the sialic acid receptors which are attached to the viral haemagglutinin, from the cell surface membrane and hence viruses cannot be released;

@ 1m [3]

(iii) Suggest a limitation of using this type of influenza drug.

1. Concentration of inhibitors may not be high enough to target all neuraminidase/ large number of host cells, difficult to ensure inhibitors reach all the host cells;
2. Antigenic drift/shift/mutation in neuraminidase gene will result in a different 3D conformation of neuraminidase, so drug needs to be continually updated;

AVP;

@ 1m [1]

(6) Describe the differences between the lysogenic life cycle of a lambda phage and the life cycle of an influenza virus.

Lysogenic life cycle	Influenza life cycle
1. Use tail tip to bind to specific receptors on bacterial cells	Use haemagglutinin to bind to sialic acid receptors on epithelial cells;
2. Only DNA is injected into host cell /Capsid left outside host cell	Capsid and viral RNA enters host cell;
3. Viral DNA integrated into host cell genome directly	Viral RNA transcribed into mRNA by RNA-dependent RNA polymerase directly;
4. Occurs in host cell bacteria	Occurs in epithelial cells;
5. Do not result in death of the host cell	Release of influenza virus by budding may result in death of host cell;

AVP;

@1m, max 2

[2]

[Total: 10m]

- 4 The most common type of lung cancer is non-small cell lung cancer (NSCLC) which affects between 75% to 85% of all lung cancer patients. The *c-Met* gene which is located on chromosome 7 has been implicated in NSCLC.

To investigate the role of *c-Met* gene in NSCLC, researchers studied the level of expression of c-Met receptor tyrosine kinase protein in several NSCLC cancer cell lines.

cDNA was generated from mRNA samples from the cancer cell lines and subsequently sequenced. Fig. 4.1a shows the normal cDNA. The corresponding protein regions it codes for are labelled. Fig. 4.1b shows two mutant cDNAs and the corresponding protein regions they code for. The corresponding regions shown in the mutant cDNA may or may not be translated to functional protein structures.

Normal cDNA:

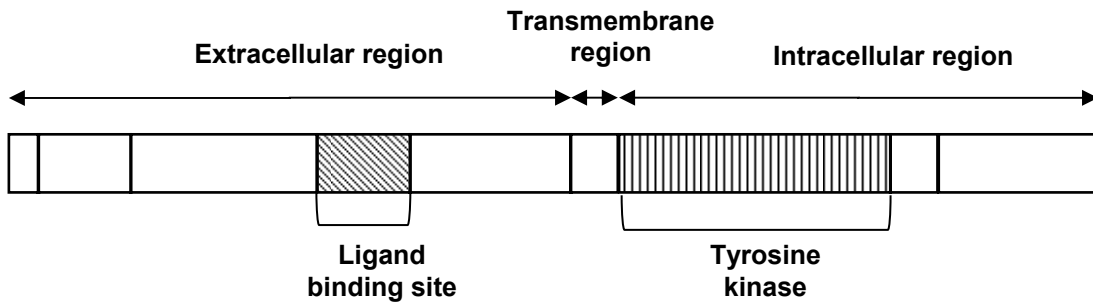


Fig. 4.1a

Mutant 1 cDNA:



Mutant 2 cDNA:



Fig. 4.1b

With reference to Fig. 4.1a and 4.1b,

- (a) (i) State and explain the types of mutation in the *c-Met* gene that could have resulted in mutant 1 and mutant 2.

Mutant 1: (max 2)

1. Insertion/deletion/substitution of nucleotide(s);
2. Such that there is the creation of a new splice site in the mRNA;
3. Spliceosome deletes one or more exons resulting in a shortened sequence coding for the extracellular region of the protein;

OR

4. Deletion of several nucleotides;
5. In the exon;
6. Hence, shorter exon resulting in a shortened sequence coding for the extracellular region of the protein;

Mutant 2: (max 2)

7. Duplication/insertion of a part of the gene sequence;
 8. Such that exons were duplicated/inserted;
 9. Resulting in a lengthened sequence coding for a part of the extracellular and transmembrane region.;
- OR
10. Insertion/deletion/substitution of nucleotide(s);
 11. Such that there is the deletion of a splice site in the mRNA;
 6. Spliceosome does not delete one or more introns resulting in a lengthened sequence coding for a part of the extracellular and transmembrane region.;

@1m, max 4

[4]

Fig. 4.2 shows the cell signaling pathway of the c-Met receptor tyrosine kinase protein.

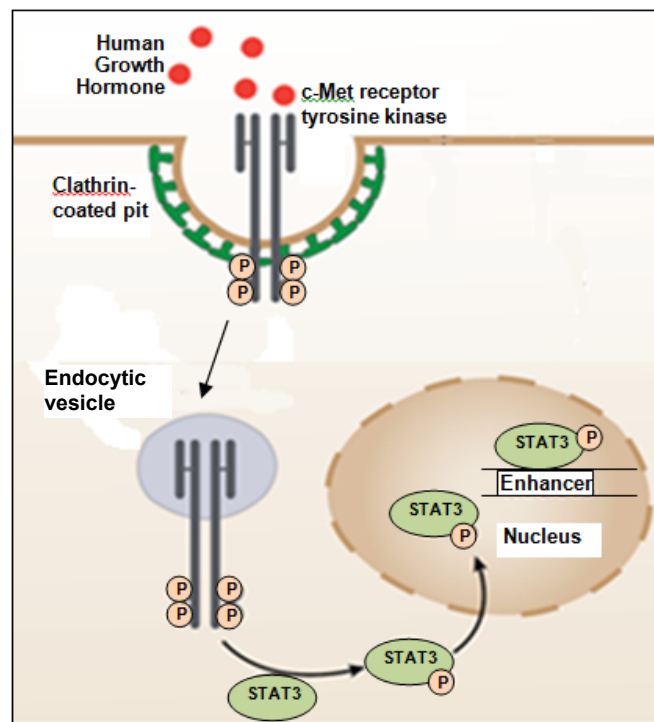


Fig. 4.2

- (6) With reference to Fig. 4.2, describe the mechanism by which STAT3 controls the expression of genes involved in cell division.

1. STAT 3 is an activator;
2. Upon phosphorylation, STAT3 changes its 3D conformation and is activated and travels into the nucleus via the nuclear pore;
3. It is complementary in conformation/configuration to the enhancer sequence and binds to it.;
4. A DNA-bending protein is recruited and causes DNA bending, (which brings STAT3 closer to the promoter.);
5. Other general transcription factors, (mediator proteins) and RNA polymerase

are also recruited to the promoter.;

6. STAT3 binds to the above, forming a transcription initiation complex.;

7. Hence, transcription of genes involved in cell division occurs at an optimal/ increased/higher rate.;

@1m, max 4

Reference to phosphorylation of STAT3 is required for awarding of full marks.

[4]

(c) Many cells are shown to have a mutation in the *c-Met* gene resulting in a hyperactive c-Met receptor tyrosine kinase. However, not all of these cells develop into cancerous cells. Explain why.

1. A loss of function mutation in both alleles of a tumour suppressor gene is also required.;

2. For the cell to divide excessively.;

3. Allowing for an accumulation of mutations in other genes involved in cancer development (e.g. telomerase, genes involved in angiogenesis etc.)

4. In a single cell lineage;

5. Cancer is a multistep process.;

@1m, max 4

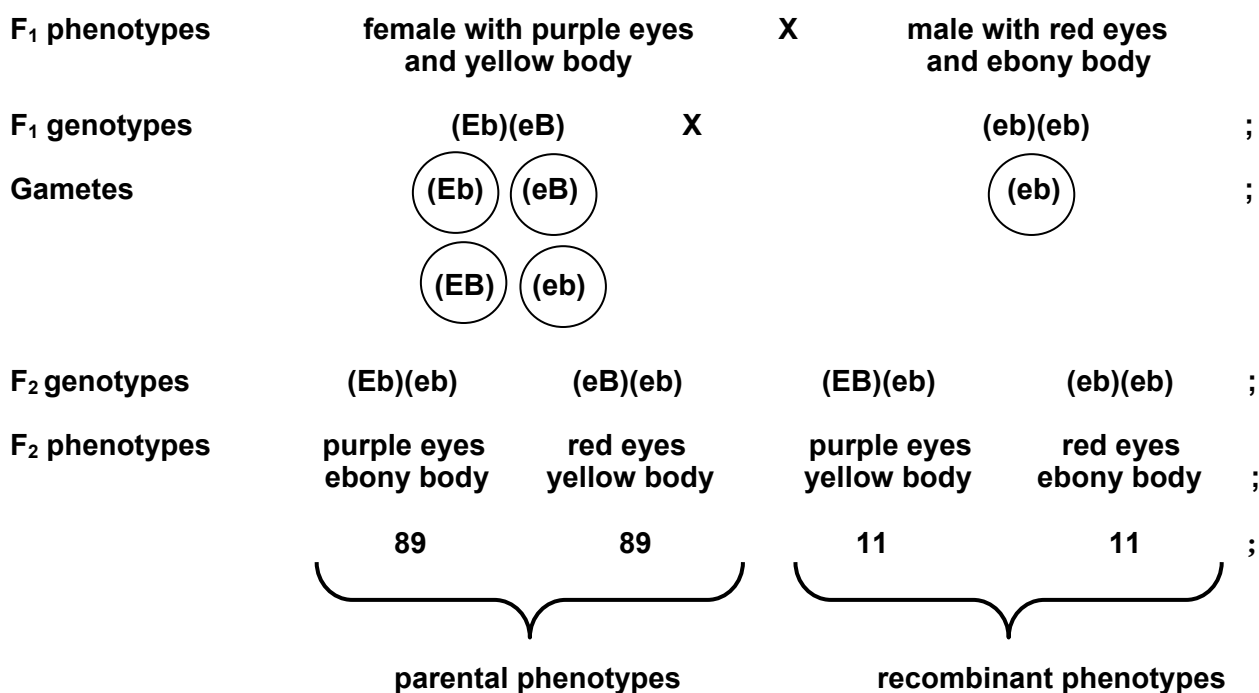
[4]

[Total: 12m]

- 5 (a) A geneticist is interested in studying eye and body pigmentation in *Drosophila*. The genes for eye colour and body colour are located 11 cM apart on chromosome 3. Purple eye is produced by a dominant mutation that is lethal in a homozygous state.

Female flies with purple eyes (**E**) and ebony bodies (**b**) were mated with male flies which were true-breeding for red eyes (**e**) and yellow bodies (**B**). All the F_1 offspring of this cross had yellow bodies. F_1 female flies which had purple eyes were chosen and test-crossed with male flies. 200 flies were present in the F_2 generation. There were equal numbers between the two parental phenotypes and equal numbers between the recombinant phenotypes.

- (i) Draw a genetic diagram of the F_1 cross to show the observed results, indicating clearly the numbers of individuals in each different phenotypic class in the F_2 offspring.



[5]

In many other crosses, the proportion of recombinant offspring obtained is lower than the one observed in (a)(i). One possible cause is the occasional occurrence of a double cross-over in female *Drosophila*, as shown in Fig. 5.1 for a doubly heterozygous female.

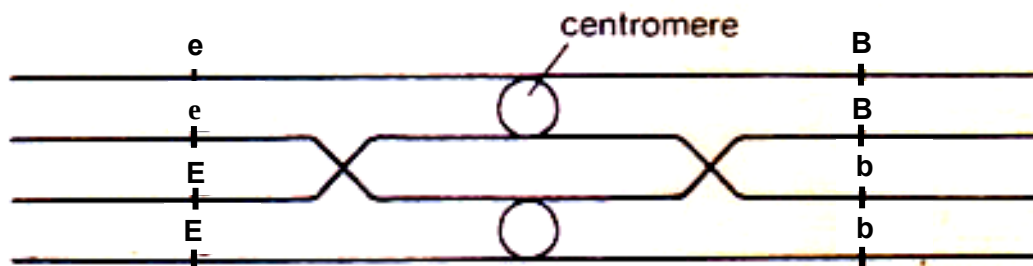


Fig. 5.1

- (ii) Assuming that a double cross-over does occur between the two genes, suggest why the recombination frequency between the two loci would be lower than the predicted percentage.

A double cross-over returns / restores the grouping of the linked alleles to the parental combination; OWTTE

[1]

In a separate experiment, the geneticist decides to investigate the chromosomal positions of genes controlling body pigmentation and wing shape in *Drosophila*. The alleles for these traits are shown below.

B: Allele for grey body **b:** Allele for black body
C: Allele for straight wings **c:** Allele for curved wings

He carried out a cross between a female fly heterozygous for both loci with a homozygous recessive black-bodied, male fly with curved wings and obtained a large number of offspring. Table 5.1 shows the results.

Table 5.1

Phenotype	Number of offspring
Grey-bodied, straight wings	35
Black-bodied, straight wings	37
Grey-bodied, curved wings	35
Black-bodied, curved wings	33

A chi-squared test was used to find out if there is any significant difference between the observed and expected ratios.

The table of probabilities is given as follows:

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- (b) The calculated χ^2 value is found to be 0.229. Using the chi-squared (χ^2) test and the information given above, state the conclusion that may be drawn from the result.

1. Calculated χ^2 value of 0.229 is less than the critical χ^2 value of 7.82;
2. At df =3;
3. p> 0.05;
4. Difference between observed ratio and expected ratio is not significant and probably due to chance;
5. *The genes controlling body pigmentation and wing shape are not linked / are found on separate chromosomes;

*compulsory pt
@ 1m [4]

[Total: 10m]

6 Fig. 6.1 shows an electron micrograph of a plant cell.



Fig. 6.1

(6) (i) Name the macromolecule which makes up structure A.

Cellulose; [1]

(ii) Describe how the structure of the named macromolecule in (a)(i) gives rise to its fibrous nature.

1. **Polymer of β glucose residues with alternate glucose residues rotated 180° ;**

2. **OH groups found on both sides of the chain enables crosslinking / formation of H bonds with other chains;**

3. **Chains associate to form microfibrils which are arranged in larger bundles to form macrofibrils (idea of bundling is important);**

4. **Chains form long strands/fibres;** @ 1m, max 3 [3]

Fig. 6.2 shows how a seed undergoes germination to form a seedling.



Fig. 6.2

(b) (i) State what is meant by respiratory quotient (RQ).

1. RQ is the ratio of carbon dioxide given out to oxygen taken in during respiration;

[1]

(ii) In some pea plants, the RQ has been found to be between three and four at the start of germination. Suggest a reason to explain the high RQ obtained at the early stages of germination.

1. The testa or seed coat still covers the seed/seed is embedded deep in the soil, making it difficult for oxygen to penetrate inside;

2. Respiration in seed is partly anaerobic (hence, the $RQ > 1$);

[2]

(6) Following root growth, leaves will develop and this is necessary for the seedling to harvest energy from the sun. Light intensity plays a role in determining the rate of photosynthesis. State why the rate of photosynthesis remains constant even at high light intensity.

Light is no longer limiting / another factor e.g. CO_2 or temperature is now limiting/

light saturation has been reached;

@ 1m [1]

[Total: 8m]

- 7** The Fluid Mosaic Membrane Model of biological membrane structures was proposed as a basic framework model to describe the basic structures of biological membranes.

Synapses are unique cell junctions found between a neurone and another cell that serve to conduct signals in a neural pathway. A chemical synapse is cholinergic if it uses acetylcholine as its neurotransmitter.

- (6)** Explain the significance of the Fluid Mosaic Model of membrane structures in the transmission of signals across a cholinergic synapse.

- 1. Fluid: phospholipids free to move within the membrane;**
- 2. allow synaptic vesicles containing neurotransmitters to fuse with plasma membrane of pre-synaptic neurone to be released into the synaptic cleft/ allow membrane proteins to change conformation even when embedded in membrane;**
- 3. Mosaic: Presence of (a variety of) proteins scattered on/in the membrane;**
- 4. Voltage-gated Ca^{2+} ion channels which allow Ca^{2+} influx across pre-synaptic membrane upon membrane depolarisation/ Chemically-gated Na^{+} ion channels which allow Na^{+} influx across post-synaptic membrane upon binding of neurotransmitters (to their specific receptors);**

[4]

Transmission at the neuromuscular junctions takes place in the same way as synapses between nerve cells. Fig. 7.1 shows a chemical synapse formed by the contact between a motor neurone and a muscle cell at the neuromuscular junction.

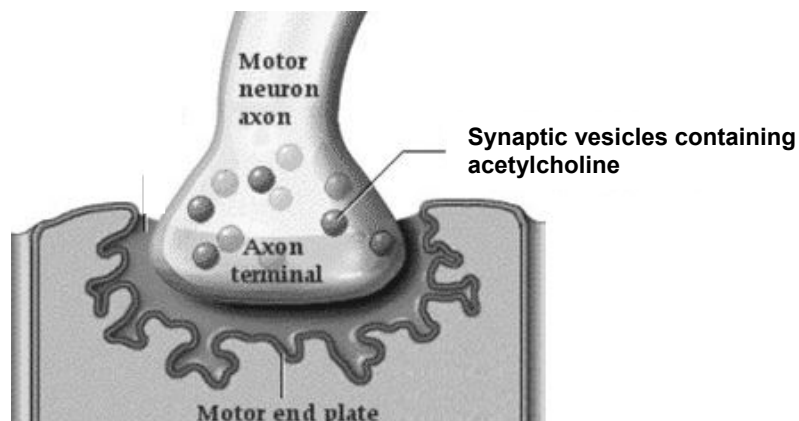


Fig. 7.1

Suxamethonium is a drug that mimics the action of acetylcholine (Ach) at acetylcholine receptors of the motor end plate. Unlike Ach, it cannot be hydrolysed by acetylcholinesterase.

- (b) With reference to the above information, explain the effect of suxamethonium on the muscles.

1. Suxamethonium is structurally similar to acetylcholine;
2. Hence, it competes for binding to the Ach receptors at the motor end plate;
(R! Competitive inhibition)
3. Because it cannot be hydrolysed by acetylcholinesterase, this results in prolonged opening of the chemically-gated Na^+ ion channels ;
4. Influx of Na^+ ions results in depolarization of the motor end-plate;
5. the end plate remains depolarised/ cannot be repolarize;
6. Muscles remained contracted/ cannot relax/ OWTTE;

[4]

[Total: 8m]

- 8 (a)** The camelids are a group of mammals that share a large numbers of homologies. Recognisable camelids have been found fossilized in rock that are 45 million years old. Modern camelids have a discontinuous distribution. Guanacos and vicunas are found in South America together with the domesticated llamas and Alpacas. Bactrian Camels and Dromedaries are found in the Asian and South African deserts.

- (i)** Explain how homologies in camelids supports Darwin's theory of descent with modification.

- 1. Presence of homologous structures in all the camelids shows that they inherited a common set of genes from common ancestor;**
- 2. Ref to anatomical, embryological and molecular homologies;**
- 3. (Modification arises due to) natural selection selecting for better adapted traits/ phenotypes in each population;**
- 4. Due to different selection pressure found in different geographical location;**

[4]

- (ii)** Without considering the theory of continental drift, explain how discontinuous distribution of modern day camelids poses a problem to Darwin's theory of descent with modification.

- 1. Different species which are descendants of a common ancestors should be found close in geographical proximity;**
- 2. As their range is limited by geographical barriers, the ocean between continents;**
- 3. Hence being found on different continents in present day suggest modern camelids may not descended from a common ancestor;**

[2]

- (b) Fig. 8.1 is a graph which shows the relationship between the number of amino acid changes of the protein haemoglobin with respect to the number of years since two organisms diverged from their common ancestor.

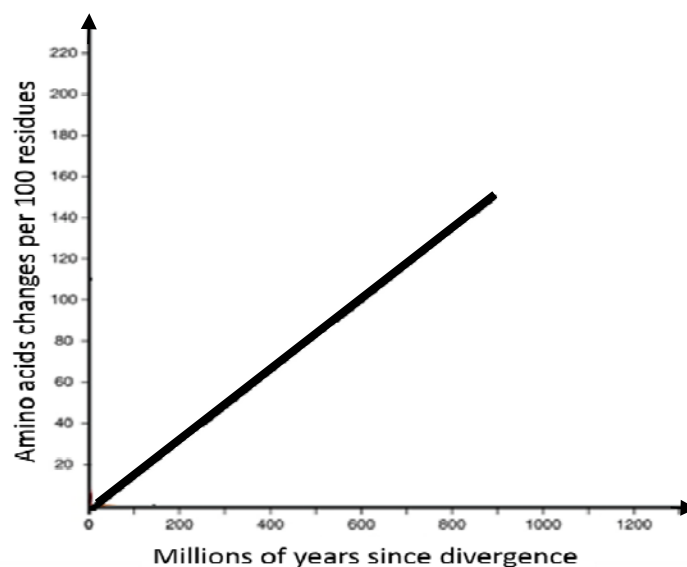


Fig. 8.1

- (i) Explain the significance of the shape of the graph shown in Fig. 8.1.

1. The linear graph shows mutation accumulates at a constant rate over time;
2. Shows that most mutation / amino acid differences between organisms are neutral mutation / silent mutation;
3. As the mutation does not come under selection pressure;
4. Frequency of the mutation in the population changes mainly due to genetic drift;
5. Elaborate genetic drift;
6. Use data as molecular clock to calculate no. of years since divergence of a species from ancestral species;

[4]

- (ii) Compare neutral mutation and silent mutation.

1. Both does not affect the fitness of the organism;
2. Both involve a change in nucleotide sequence;
3. However silent mutation does not involve a change to the amino acid sequence of the polypeptide;
4. While neutral mutation includes changes to the amino acid sequence;
5. Both do not change the phenotype of an organism;

[3]

[Total: 13m]

Section BAnswer **EITHER 9 OR 10.**

Write your answers in the lined pages provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

EITHER

9 (a) Using the induced-fit hypothesis, explain the mode of action of enzymes. [6]

1. Enzyme lowers activation energy/ alternative pathway of lower activation energy;
2. Ref. to *mechanisms* e.g. 'proximity effect', 'strain effect' and 'orientation surface' (any one)
3. Enzyme specific in its action;
4. due to complementary 3D configuration/conformation of active site to that of substrate;
5. The induced fit model suggests that the enzyme and the substrate do not fit together exactly;
6. Effective collisions between enzymes and (specific) substrate molecules result in substrate binding to active site of enzyme;
7. The enzyme undergoes a 3D conformation change, which improves the fit between substrate and enzyme;
8. to form enzyme-substrate (ES) complexes;
9. Product formed that no longer fits into active site and is released;
10. Enzyme remains unchanged at the end of the reaction (and can be reused);

- (b) With reference to haemoglobin, explain the significance of bonds in maintaining the protein's structure and function. [8]

Compulsory points (must be seen for full marks):

1. The primary structure is maintained by peptide bonds between amine groups and carboxyl groups of amino acids;
2. The secondary structure which refers to the α helix is maintained by hydrogen bonds between $-\text{CO}$ and $-\text{NH}$ groups of the polypeptide backbone;
3. The tertiary structure which refers to the overall 3D configuration/ globular shape of the subunit that is maintained by hydrogen bonds, ionic bonds and hydrophobic interactions (any 2) between the R groups of amino acid residues.;
4. The quaternary structure which refers to the association of the two polypeptide chains in **each dimer** and association of the **two dimers** are held together by mainly **hydrophobic interactions** and **weak hydrogen and ionic bonds** respectively; (give for quaternary structure if student describes either association within or between dimers)

N.B. If there is none of the above points, max 4 marks can be awarded.

Description of tertiary structure:

5. Globular structure allows for the packing of many haemoglobin in red blood cells;
6. Each globin polypeptide is folded such that the bulk of the hydrophobic amino acid residues are buried in the interior of the globular structure;
7. Ref. to haem binding pocket lined with hydrophobic amino acids to provide a hydrophobic environment for hydrophobic haem group to bind;
8. Ref. to presence of Fe^{2+} to allow for reversible binding of O_2 ;
9. Hydrophilic amino acid residues are on the outside (to form hydrogen bonds with water);
10. Haemoglobin is soluble in aqueous medium and hence a good transport protein for oxygen in blood;

Description of quaternary structure:

11. (Association of two dimers with weak hydrogen and ionic bonds) results in the ability of the two dimers to move with respect to each other;
12. This allows for cooperativity;
13. When an oxygen molecule binds to/is released from 1 haemoglobin subunit, the binding/ release induces a conformational change in the remaining subunit;
14. Which increases/ lowers the affinity for oxygen of the remaining three oxygen binding sites respectively;
15. Hence facilitates the loading and unloading of oxygen;

(c) Compare competitive and non-competitive inhibition of enzyme action.

[6]

Similarities:

1. At low substrate concentration, rate of reaction in the presence of inhibitors is slower than that in the absence of inhibitor;

Features	Competitive	Non-competitive
Structure of inhibitor	2. Resembles substrate;	Does not resemble substrate;
Binding site of inhibitor	3. Binds to active site of enzyme;	Binds to enzyme at a region other than the active site/ allosteric site;
Mechanism of inhibition	4. Blocks substrates from binding to active site of the enzyme;	Blocks substrates from binding to active site by changing the conformation of the active site;
Effect of high substrate concentration on inhibition	5. Inhibition can be reversed at high substrate concentration; 6. $V_{\max}$ in the presence of inhibitor can be very close to that of reaction in the absence of inhibitor;	Inhibition cannot be reversed at high substrate concentration; $V_{\max}$ in the presence of inhibitor is less than that of reaction in the absence of inhibitor;

[Total: 20m]

OR

10 (a) Explain the roles of the key components of the homeostatic control system. . [6]

1. It consists of the following components: detector, control centre and effector;
2. The detector/ receptor senses the stimuli and the information is relayed to the control centre;
3. The control centre (receives the information and) compares it with the set-point of that parameter;
4. Which is the optimal level of that specific parameter;
5. When there is a deviation in the parameter from the set point, the control centre initiates an appropriate response *(to restore conditions back to the set point);
6. This action is conveyed to the effector through the communication systems/nervous & endocrine systems in the body;
7. The effector will carry out the response (initiated by the control centre to restore conditions to the set point);
8. Ref to negative feedback mechanism;

* Ref to be made at least once in answer
@ 1m, max 6

(b) With the use of named examples, describe the role of hormones in the regulation of blood glucose concentration in humans. [8]

Insulin:

1. When blood glucose concentration is higher than the set point of 80-90mg/100ml;
2. The change is detected by β cells (α and β cells of the islets of Langerhans) and are stimulated to secrete (more) insulin (and secretion of glucagon by α cells is inhibited);
3. Insulin binds to cell-surface insulin receptors on the plasma membrane of effector cells e.g. liver, skeletal muscles and adipose cells (at least one);
4. (Upon binding of insulin,) the insulin receptor, which is a receptor tyrosine kinase, phosphorylates intracellular enzymes to bring about the following effects::;
5. Acceleration of the rate of glucose uptake via facilitated diffusion;
6. By increasing the number of glucose transporters in the plasma membrane of cells (but not in liver cells);
7. Rate of glycolysis is increased, glycogenesis is stimulated, glycogenolysis is inhibited, stimulates amino acid absorption and protein synthesis, inhibits gluconeogenesis, stimulates lipogenesis.; (any two)

[max 4 marks]

Glucagon:

8. When the blood glucose concentration is lower than the set point of 80 - 90mg/100ml;
9. The change is detected by α cells (α and β cells of the islets of Langerhans) and are stimulated to secrete (more) glucagon (and secretion of insulin by β cells is inhibited);
10. Glucagon binds to the cell-surface glucagon receptors on the plasma membrane of effector cells e.g. liver;
11. Upon binding of glucagon to its receptor, a G-protein coupled receptor, adenylate cyclase is activated to produce cyclic AMP (cAMP) from ATP to bring about the following effects;;
12. Stimulates glycogenolysis, gluconeogenesis and lipolysis.; (any two)
[max 4 marks]
13. Hence, the opposite actions of both hormones help to maintain the blood glucose concentration about a set point;
14. The ratio of the amounts of both hormones determines the net effect.;

(c) Explain the significance of cell signalling pathways in homeostasis.

[6]

1. Cell signaling pathways allows for cellular response by an extracellular signal.;
2. Such that a constant internal environment can be maintained.;
3. It allows for signal amplification;
4. Since at each catalytic step in the cascade, the number of activated products is much greater than in the preceding step.;
5. A small number of extracellular signal molecules is sufficient to elicit a large cellular response.;
6. Ref to a small number of glucagon molecules binding to receptors on the surface of an effector cell can lead to the release of hundreds of millions of glucose molecule from glycogen.;

[max 2]

7. It allows for coordination and regulation (to information coming from different sources in a body).;
8. A pathway that is triggered by a single kind of signal can diverge to produce two or more cellular responses.;
9. Ref. to receptor tyrosine kinases which can activate multiple relay proteins or second messengers which can regulate numerous proteins.;
10. Different signal transduction pathways can also converge to modulate a signal response;

[max 2]

11. It contributes to the specificity of the response;
12. Because of the specific constitution of proteins within target cell such as signalling proteins, receptors, relay proteins to carry out the response in a particular cell type.;
13. A specific signal molecule must bind to a specific receptor with a specific complementary conformation to elicit a specific reaction via a specific pathway;
14. Ref. to muscle cells responding to insulin but not glucagon;
15. Hence, two cells can respond differently to the same signal / Multiple responses in different cells to one signal molecule because 1 ligand can trigger multiple signal transduction pathways to elicit different responses;
16. Ref. to Insulin causes increase in glucose uptake in muscle and fat cells but not in liver cells because liver cells do not express a specific type of glucose transporter.;

[max 2]

17. The hormone may be carried in the blood stream and cells in all parts of the body will be exposed to the hormone;
18. Hence, as long as the cells have the appropriate receptor, many cells in different parts of the body can be activated and they respond simultaneously;
19. Ref. to Insulin causes increase in glucose uptake in muscle and fat cells in different parts of the body;

[max 2]

N.B. Students have to discuss at least 3 of the 4 areas to be awarded the full marks.

[Total: 20m]